

B.E. II EXAMINATION DEC.' 2023
E&TC Engineering
3ATRC1: Applied Mathematics-III

Time: 3 Hrs.]

[Max. Marks: 60]

Note: Attempt any TWO parts from each Question. Each carries equal marks.

Q.1. (a) Find the Nature of Singularities for the following functions. (06)

i) $f(z) = \frac{\sin z}{z}$ ii) $g(z) = \frac{e^{1/z}}{z^2}$ iii) $u(z) = \frac{e^{2z}}{(z-1)^4}$

(b) If $f(z_1) = \oint_C \frac{3z^2+7z+1}{z-z_1} dz$ where C is the circle $x^2 + y^2 = 4$, find the value of (06)

(i) $f(3)$, (ii) $f'(1-i)$, (iii) $f''(1-i)$.

(c) Find Laurent expansion for $f(z) = \frac{7z-2}{z^3-z^2-2z}$ in the region given by (06)

i) $0 < |z+1| < 1$, ii) $1 < |z+1| < 3$, iii) $|z+1| > 3$.

Q.2. (a) The transition matrix P of the markov chain is as, (06)

$$P = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Initial probability distribution $P(0) = [2/3, 0, 1/3]$, if p_{ij} denotes m^{th} step transition probability, find $P^{(4)} \cdot P_{13}^{(2)}$, $P_{32}^{(2)}$.

(b) An electronic system comprises 6 silicon transistors, 4 germanium diodes, 8 film resistors, and 3 tantalum capacitors connected in a parallel circuit. Each component has its hourly failure rate: (06)

1. Silicon transistor: $\lambda = 6 \times 10^{-5}$
2. Germanium diodes: $\lambda = 2 \times 10^{-5}$
3. Film resistors: $\lambda = 1.5 \times 10^{-4}$
4. Tantalum capacitors: $\lambda = 1.8 \times 10^{-4}$

Calculate the reliability of the circuit for 10 hours, when the components follow exponential distribution.

(c) A Traveler visits one of three cities X, Y, and Z every day, never visiting the same city on consecutive days. If the traveler visits city X, the next day they visit city Y. If the traveler visits either city Y or Z, the next day they are twice as likely to visit city X as to visit the other city. If the traveler visits city Z on a Monday, what is the probability that they visit city X on the following Friday? For X, Y and Z: (06)

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 2/3 & 0 & 1/3 \\ 2/3 & 1/3 & 0 \end{bmatrix}$$

Q.3. (a) (i) Evaluate $\int_0^4 e^x dx$ by Simpson's rule given that (06)

$e = 2.72$, $e^2 = 7.39$, $e^3 = 20.09$, $e^4 = 54.6$.

(ii) Use Trapezoidal rule to evaluate $\int_0^1 x^3 dx$ considering 5 sub-intervals.(b) Use Stirling's formula to evaluate $f(32)$ given that (06)

x	20	25	30	35	40
F(x)	14.035	13.674	13.257	12.089	11.309

- (c) Using Newton's divided difference formula, find $f(1)$. (06)

x:	-4	-1	0	2	5
F(x):	1245	33	5	9	1335

- Q.4. (a) Solve the system of equations by Gauss-Seidel iterative method correct to 2 decimal places. (06)

$$\begin{aligned} 54x + y + z &= 110, \\ 2x + 15y + 6z &= 72, \\ -x + 6y + 27z &= 85. \end{aligned}$$

- (b) Apply Runge-Kutta method (fourth order), to find an approximate value of y when $x = 2.0$, given that $\frac{dy}{dx} = x + y^2$ and $y(0) = 1$. [Interval $h = 0.1$] (06)

- (c) Find real root of equation $x^4 - x = 9$, by Newton-Raphson method correct to 3 decimal places. (06)

- Q.5. (a) (i) Write Dirichlet's conditions for Fourier series. (03)
(ii) Express the function $f(x)$ as Fourier integral. (03)

$$f(x) = \begin{cases} 1, & |x| \leq 1 \\ 0, & |x| > 1 \end{cases} \quad (03)$$

- (b) Find the Fourier half-range cosine series of the function (06)
- $$f(x) = \begin{cases} 2t, & 0 < t < 1 \\ 2(2 - t), & 1 < t < 2 \end{cases}$$

- (c) Find the Fourier series corresponding to the function $f(x)$ defined in $(-2, 2)$ as follows (06)

$$f(x) = \begin{cases} 2, & -2 \leq x \leq 0 \\ x, & 0 < x < 2 \end{cases}$$
